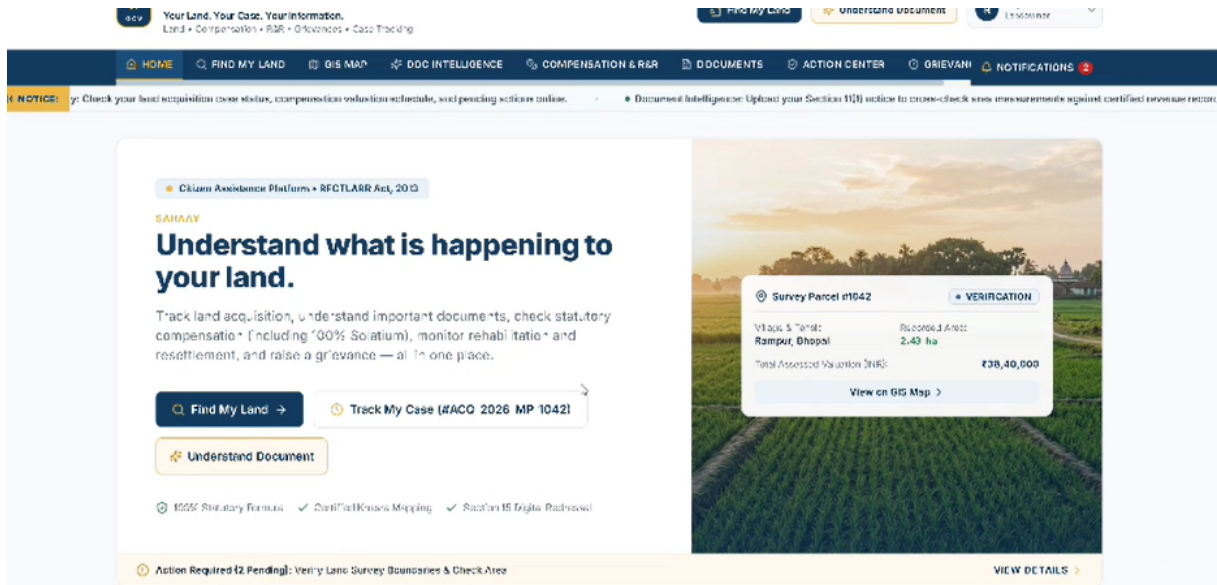


SAHAAY

Citizen-First Land Acquisition & Compensation Companion

TECHNICAL PROJECT REPORT



Team Pioneers | Mission 001 Submission

1. Abstract

Sahaay is a citizen-first digital platform designed to simplify land acquisition information for rural landowners, land-affected families, and citizens with limited digital literacy. It combines document intelligence, GIS visualization, compensation tracking, grievance management, and transparent status monitoring into a single accessible experience.

The platform bridges the information gap between citizens and administrative processes by presenting acquisition-related information in a simplified and understandable manner. It also provides an officer-facing workspace for administrative operations, connecting the citizen and administrative sides of the workflow.

2. Introduction

2.1 Background & Motivation

Land acquisition involves land records, parcels, notices, project information, compensation, rehabilitation-related information, documents, and administrative actions. When these elements are difficult to understand or distributed across different processes, citizens can struggle to determine what is happening to their land and what action is required.

2.2 Objectives

- Centralize land-acquisition information in a single digital interface.
- Provide citizen-friendly access to land, case, document, compensation, and grievance information.
- Visualize relevant land parcels through GIS-based mapping.
- Assist with document information extraction and potential discrepancy detection.
- Enable transparent case and grievance status tracking.
- Provide an officer-facing workspace for administrative operations.

2.3 Mission 001 Alignment

Sahaay addresses a public-utility problem for an underserved user group. It focuses on accessibility, transparency, reduced information barriers, and automation through document intelligence.

3. Problem Statement & Existing Process

Citizens often receive complex acquisition notices and struggle to understand compensation, project progress, rehabilitation benefits, and grievance procedures. Information can be fragmented and difficult to access, while legal terminology and administrative workflows can be difficult for non-specialist users.



3.1 Key Gaps

- Fragmented access to acquisition-related information.
- Complex terminology and procedures.
- Difficulty understanding affected land geographically.
- Limited visibility into case and compensation status.
- Manual effort for lengthy documents.
- Difficulty identifying information discrepancies.

4. Proposed System — Sahaay

Sahaay brings the citizen and administrative sides of the process into one digital platform. The citizen interface focuses on understanding, tracking, and acting; the officer interface focuses on reviewing and managing operational information.

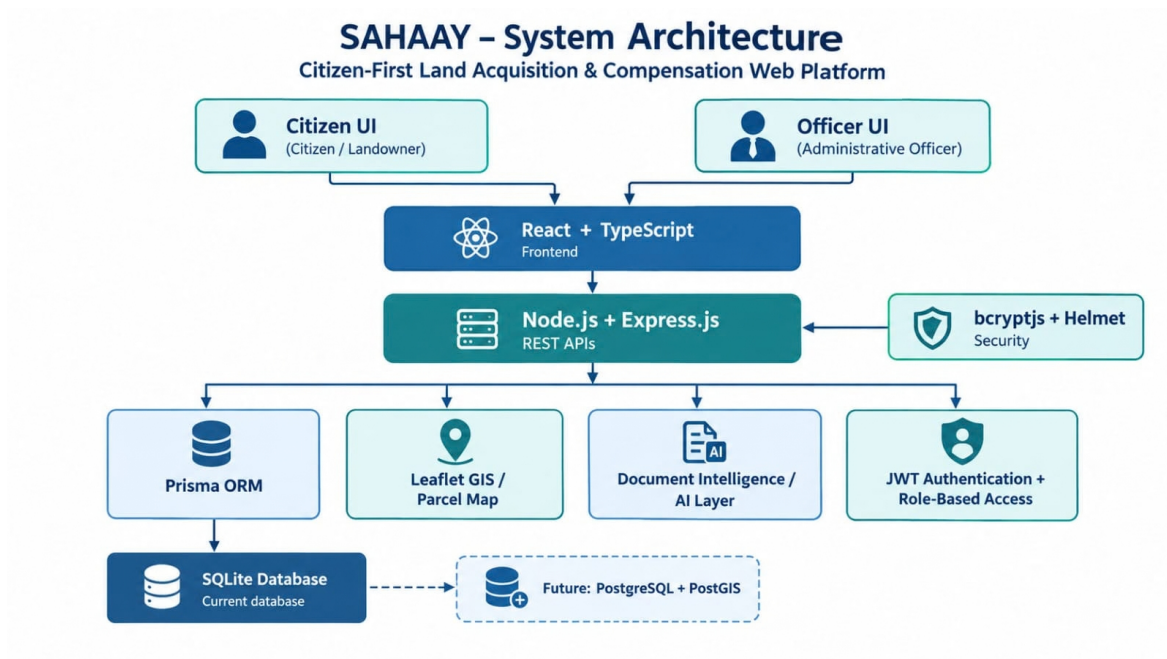
User	Core Capabilities
Citizen	Land/case information, GIS parcel view, documents, compensation, grievances, status tracking.
Officer	Administrative workspace, case review, documents, grievances, operational/status updates.

4.1 Core Modules

- Authentication & role-based access
- Citizen dashboard
- Officer workspace
- Land/parcel information
- GIS visualization
- Document intelligence
- Compensation tracking
- Grievance management
- Notifications/activity tracking

5. System Architecture

The current implementation uses a modular web architecture. React and TypeScript provide the frontend; Node.js and Express provide the backend; Prisma provides database access; SQLite is the current database; and Leaflet provides GIS visualization. PostgreSQL/PostGIS is identified as a future scaling path.



5.1 Technology Stack

Layer	Technology	Purpose
Frontend	React + TypeScript	Web interface
Styling	Tailwind CSS	UI styling
Backend	Node.js + Express.js	API/application layer
ORM	Prisma	Database abstraction
Database	SQLite	Current persistence
GIS	Leaflet	Map/parcel visualization
Authentication	JWT	Authentication/authorization
Security	bcryptjs + Helmet	Password hashing / HTTP security

5.2 Architecture Principles

- Modularity and maintainability.
- Role separation between citizen and officer experiences.
- Scalability readiness for PostgreSQL/PostGIS.
- Traceability through audit mechanisms.

6. Functional & Non-Functional Requirements

ID	Functional Requirement
FR-01	User authentication and role distinction.
FR-02	Centralized acquisition-related information.
FR-03	GIS-based land/parcel visualization.
FR-04	Document information extraction.
FR-05	Discrepancy detection within the implemented workflow.
FR-06	Case/acquisition monitoring.
FR-07	Citizen grievance submission and tracking.
FR-08	Officer administrative workspace.
FR-09	Notifications/activity records where implemented.

6.1 Non-Functional Requirements

Security: authentication, authorization, password hashing, validation and audit logging.

Usability: simple information hierarchy and citizen-oriented presentation.

Maintainability: modular architecture and ORM-based access.

Scalability: future PostgreSQL/PostGIS and integrations.

Accessibility: mobile-first and simplified presentation.

7. Database Design

Entity	Purpose
User	Authentication
CitizenProfile	Citizen details
Project	Infrastructure projects
Parcel	Land parcel records
AcquisitionCase	Case management
CompensationRecord	Compensation tracking
Grievance	Complaint handling
Notification	Alerts/updates
AuditLog	Activity tracking

8. System Workflows

8.1 Citizen Workflow

Login → Dashboard → Find Land/Case → Parcel → Timeline → Documents → Compensation → Grievance → Status

The citizen journey is organized around the information needs of a person affected by land acquisition, moving from identification of the relevant case or parcel to understandable status, document, compensation, and grievance information.

8.2 Officer Workflow

Officer Login → Workspace → Cases → Select Case → Review Land/Documents → Grievances → Update Case/Status

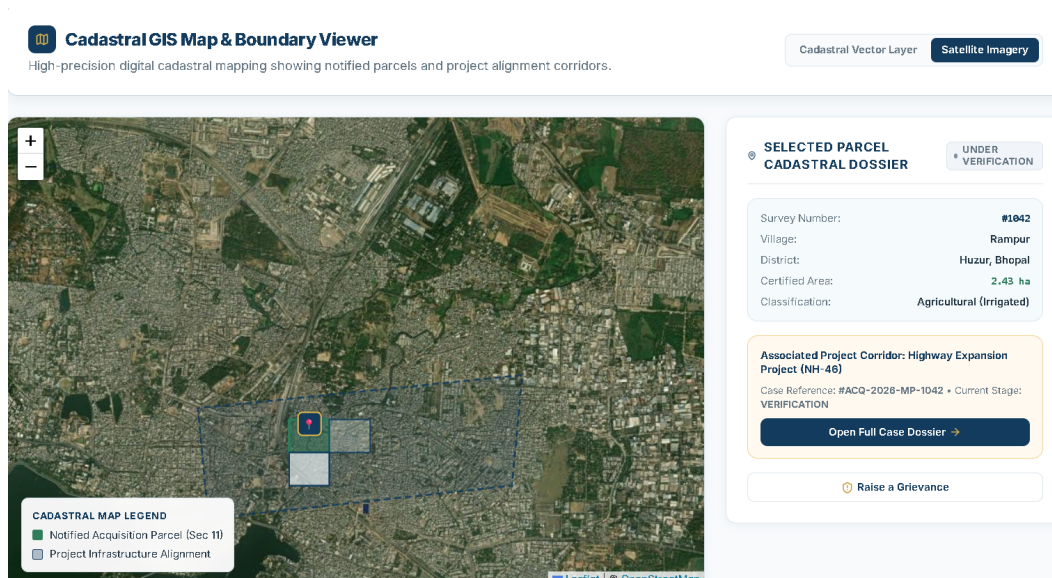
The officer workspace centralizes operational information for case and grievance handling.

8.3 Citizen-Officer Loop

Citizen action → Sahaay workflow → Officer review/action → Updated status → Citizen

9. GIS / Land Parcel Visualization

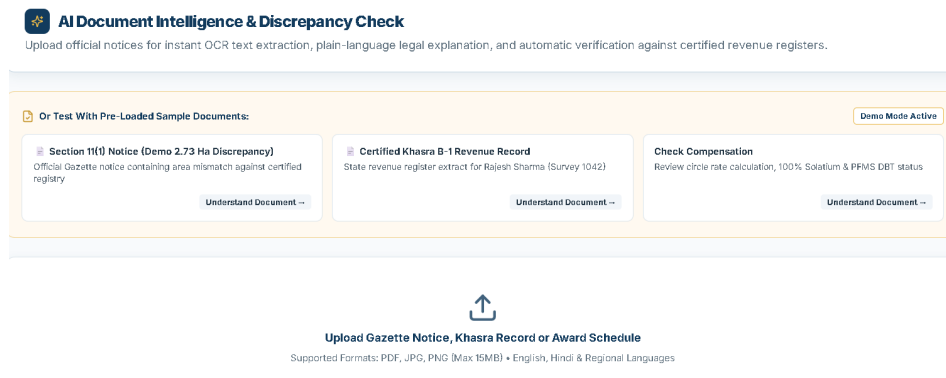
GIS connects administrative land information with geographic context. Sahaay uses Leaflet for map-based visualization so relevant parcel and location information can be presented in an intuitive form.



- Visual understanding of land information.
- Geographic context for parcels.
- Foundation for future cadastral/spatial-data integrations.

10. Document Intelligence

Sahaay uses a deterministic AI architecture for document understanding, structured extraction, discrepancy detection, and simplified explanations. The goal is to reduce the effort required to manually review complex administrative documents.



11. Compensation & Grievance Management

11.1 Compensation Tracking

Compensation tracking provides visibility into compensation-related information associated with an acquisition case. CompensationRecord is included as a dedicated database entity.

11.2 Grievance Management

The grievance system provides a structured mechanism for complaint handling. Citizens can raise issues and track their status, while the officer workspace supports administrative handling.

12. Security & Privacy

Control	Purpose
JWT	Authentication and authorized access
Role-Based Access	Citizen/officer separation
bcryptjs	Password hashing
Validation	Input protection
Helmet	Security-related HTTP headers
Audit Logging	Action traceability

13. User Interface & UX

The interface is designed around a citizen-first experience with a separate officer workspace. The visual language connects Sahaay with land records, mapping, land administration, and transparent case management.



14. API & Backend Design

The backend uses Node.js and Express.js to provide application services to the React frontend, with Prisma for database access. Final documentation should contain only endpoints present in the implementation.

Method	Endpoint	Purpose	Access
POST	[actual login endpoint]	Authentication	Public
GET	[actual land/case endpoint]	Retrieve case information	Authorized
POST	[actual document endpoint]	Upload/process document	Authorized
POST	[actual grievance endpoint]	Create grievance	Citizen
GET/PUT	[actual officer endpoints]	Manage cases/status	Officer

15. Testing

Test	Expected Result
Citizen login	Citizen dashboard opens
Officer login	Officer workspace opens
Invalid login	Error shown
Land/case search	Relevant data displayed
GIS view	Map loads
Document upload	Processing workflow starts
Grievance	Grievance recorded/tracked
Role restriction	Unauthorized access blocked

16. Results & Impact

Sahaay increases visibility and accessibility for citizens affected by land acquisition while providing an administrative workspace for officers. Its value comes from connecting land information, maps, documents, compensation, grievances, and status tracking around one workflow.

Citizen-Side Value	Officer-Side Value
Centralized information	Centralized administrative workspace
Parcel visualization	Case review
Document/compensation visibility	Document and grievance review
Grievance tracking	Status management
Case visibility	Activity/audit traceability where implemented

17. Limitations

- Current database uses SQLite; PostgreSQL/PostGIS is future scope.
- Document processing depends on document quality and implemented extraction workflow.
- Government-system integrations are future scope unless explicitly implemented.
- GIS depends on available spatial/parcel data.
- Production deployment requires additional operational, security, and compliance validation.

18. Future Scope

- PostgreSQL/PostGIS migration.
- Multilingual support.
- Advanced OCR/document understanding.
- Authorized government-system integration.
- Mobile application.
- Expanded cadastral/spatial visualization.
- Additional citizen notification channels.

19. Conclusion

Sahaay demonstrates how technology can transform complex land-acquisition processes into a more understandable citizen experience while also supporting administrative operations. By combining GIS visualization, document intelligence, compensation tracking, grievance management, role-based access, and status monitoring, Sahaay provides a modular foundation for a more connected digital land-acquisition experience.

20. References

1. Department of Land Resources, Government of India
2. DILRMP documentation used for domain understanding.
3. React / TypeScript documentation.
4. Node.js / Express.js documentation.
5. Prisma documentation.
6. Leaflet documentation.
7. Documentation for the AI/document-processing technology actually used.